

TaniCare AI – Kawan Petani, Penjaga Tanaman

Team: Jexko-1Bit
Hackathon: MyAI Future Hackathon 2026
Track: Track 1 – Padi & Plates (Agrotech & Food Security)

Project Overview

TaniCare AI is a smart mobile application designed to help Malaysian smallholder farmers detect crop diseases instantly and receive intelligent recommendations using Google AI technologies.

By combining **Gemini 1.5 Pro** for image analysis, **Google Earth Engine** for advanced satellite monitoring (NDVI + EVI + anomaly detection), and **real-time weather data**, the app provides actionable insights in simple Bahasa Melayu to reduce yield losses and support national food security goals.

Key Features

- **Instant Disease Diagnosis:** Take a photo of any plant → Gemini AI instantly identifies disease, severity, treatment (organic & chemical), and prevention tips.
- **Advanced Satellite Analytics:** Real NDVI, EVI, and anomaly detection using Sentinel-2 data from Google Earth Engine.
- **Real-time Weather Integration:** Combines satellite health data with current temperature, humidity, and rainfall for smarter risk alerts.
- **Beautiful Analytics Dashboard:** Interactive line charts for NDVI trends and bar comparison between NDVI vs EVI.
- **Multi-State Support:** Works across Malaysia (Johor, Kedah, Perak, Selangor, Pahang, etc.).
- **Data Persistence:** All scans and analytics automatically saved to **Google Cloud Firestore**.
- **Farmer-Friendly UI:** Clean, modern design with intuitive bottom navigation.

Technology Stack

- **Frontend:** Flutter (Material 3) + Riverpod 2.0 (advanced state management)
- **AI Analysis:** Google Gemini 1.5 Pro (multimodal)
- **Satellite Data:** Google Earth Engine (Sentinel-2)
- **Weather Data:** Open-Meteo Real-time API
- **Database:** Google Cloud Firestore (GCP)
- **Backend:** Google Cloud Functions (Python)
- **Charts:** fl_chart
- **Performance:** Optimized with const constructors, caching, and minimal rebuilds

Project Structure

tani_care_ai/ ├── lib/ | ├── main.dart | ├── providers/ | ├── screens/ | ├── utils/constants.dart | └──
gemini_service.dart, earth_engine_service.dart ├── backend/functions/earth-engine-alert/ | ├── main.py | └──
requirements.txt ├── pubspec.yaml | └── deploy.sh └── README.md

How to Run

Flutter App

```
flutter pub get
```

```
flutter run
```

```
cd backend/functions/earth-engine-alert
```

```
./deploy.sh
```

Impact

Helps reduce crop yield losses by 30-50% through early detection

Directly supports DKMN 2030 and NAP 2.0 (National Food Security Policy)

Empowers smallholder farmers with accessible AI technology

Contributes to lowering Malaysia's food import dependency

Slogan:

"Satu Gambar, Banyak Harapan untuk Ladang Anda"

Developed by Jexko-1Bit

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